Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The percentage of Se in peroxidase anhydrous enzyme is 0.5% by weight (atomic weight $= 78.4$). Then minimum molecular weight of peroxidase anhydrous enzyme is

(A) 1.568×10^3

(B) 1.568×10^4

(C) 15.68

(D) 3.136×10^4

Q.2

The number of water molecules present in a drop of water (volume $0.0018\ \mathrm{mL}$) density $= 1\mathrm{g\ mL}^{-1}$ at room temperature is

(A) 1.084×10^{18}

(B) 6.023×10^{19}

(C) 4.84×10^{17}

(D) 6.023×10^{23}

Q.3

A mixture of O_2 and gas "Y" mol. mass 80 in the mole ratio a : b has a mean molecular mass 40 What would be mean molecular mass, if the gases are mixed in the ratio b : a under, identical conditions? (Assume that gases are non-reacting):

(A) 40

(B) 48

(C) 62

(D) 72

Q.4

If 3.01×10^{20} molecules are removed from $98\mathrm{mg}$ of $\mathrm{H_2SO_4}$, then the number of moles of $\mathrm{H_2SO_4}$ left are

(A) 0.1×10^{-3}

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(B) 0.5×10^{-3}

(C) 1.66×10^{-3}

(D) 9.95×10^{-2}

Q.5

Arrange the following in the order of increasing mass (atomic mass: $\mathrm{O} = 16, \mathrm{Cu} = 63, \ \mathrm{N} = 14$)
I. one atom of oxygen
II. one atom of nitrogen
III. 1×10^{-10} mole of oxygen
IV. 1×10^{-10} mole of copper

(A) $\mathrm{II} < \mathrm{I} < \mathrm{III} < \mathrm{IV}$

(B) $\mathrm{I} < \mathrm{II} < \mathrm{III} < \mathrm{IV}$

(C) $\mathrm{III} < \mathrm{II} < \mathrm{IV} < \mathrm{I}$

(D) $\mathrm{IV} < \mathrm{II} < \mathrm{III} < \mathrm{I}$

Q.6

Complete combustion of $0.858\ \mathrm{g}$ of compound X gives $2.64\ \mathrm{g}$ of $\mathrm{CO_2}$ and $1.26\ \mathrm{g}$ of $\mathrm{H_2O}$. The lowest molecular mass X can have:

(A) $43\ \mathrm{g}$

(B) $86\ \mathrm{g}$

(C) $129\ \mathrm{g}$

(D) $172\ \mathrm{g}$

Q.7

When burnt in air, $14.0\ \mathrm{g}$ mixture of carbon and sulphur gives a mixture of $\mathrm{CO_2}$ and $\mathrm{SO_2}$ in the volume ratio of $2:1$, volume being measured at the same conditions of temperature and pressure moles of carbon in the mixture is

(A) 0.75

(B) 0.5

(C) 0.40

(D) 0.25

Q.8

$1.12\ \mathrm{mL}$ of a gas is produced at S.T.P. by the action of $4.12\mathrm{mg}$ of alcohol ROH with methyl magnesium Iodide. The molecular mass of alcohol is

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(A) 16.0

(B) 41.2

(C) 82.4

(D) 156.0

Q.9

How many moles of P_4 can be produced by reaction of 0.10 moles $Ca_5(PO_4)_3F$, 0.36 moles SiO_2 and 0.90 moles C according to the following reaction?

$$4Ca_5(PO_4)_3F + 18SiO_2 + 30C \longrightarrow 3P_4 + 2CaF_2 + 18CaSiO_3 + 30CO$$

(A) 0.060

(B) 0.030

(C) 0.045

(D) 0.075

Q.10

1 mole of mixture of CO and CO_2 requires exactly $28\ gKOH$ in solution for complete conversion of all the CO_2 into K_2CO_3. How much amount more of KOH will be required for conversion into K_2CO_3 if one mole of mixture is completely oxidized to CO_2.

(A) $112\ g$

(B) $84\ g$

(C) $56\ g$

(D) $28\ g$

Q.11

Number of moles of $KMnO_4$ required to oxidize one mole of $Fe(C_2O_4)$ in acidic medium is

(A) 0.167

(B) 0.6

(C) 0.2

(D) 0.4

Q.12

Haemoglobin contains 0.33% of iron by weight. The molecular weight of haemoglobin is approximately 67200 . The number of iron atoms (at. wt. of $Fe = 56$) present in one molecule of haemoglobin is

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(A) 6

(B) 1

(C) 2

(D) 4

Q.13

A $25.0\ \text{mm} \times 40.0\ \text{mm}$ piece of gold foil is $0.25\ \text{mm}$ thick. The density of gold is $19.32\ \text{g/cm}^3$. How many gold atoms are in the sheet? (Atomic weight : $\text{Au} = 197.0$)

(A) 7.7×10^{23}

(B) 1.5×10^{23}

(C) 4.3×10^{21}

(D) 1.47×10^{22}

Q.14

$25.4\ \text{g}$ of I_2 and $14.2\ \text{g}$ of Cl_2 are made to react completely to yield a mixture of ICl and ICl_3 Calculate moles of ICl and ICl_3 formed

(A) $0.1, 0.1$

(B) $0.2, 0.2$

(C) $0.1, 0.2$

(D) $0.2, 0.1$

Q.15

The largest number of molecules is in

(A) $36\ \text{g}$ of water

(B) $28\ \text{g}$ of carbon monoxide

(C) $46\ \text{g}$ of ethyl alcohol

(D) $54\ \text{g}$ of nitrogen pentoxide

Q.16

$2\ \text{g}$ of oxygen contains number of atoms equal to that in

(A) $0.5\ \text{g}$ of hydrogen

(B) $4\ \text{g}$ of sulphur

(C) 7 g of nitrogen

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(D) 2.3 g of sodium

Q.17

If we consider that $1/6$, in place of $1/12$, mass of carbon atom is taken to be the relative atomic mass unit, the mass of one
mole of the substance will

(A) be a function of the molecular mass of the substance

(B) remain unchanged

(C) increase two fold

(D) decrease twice

Q.18

How many moles of magnesium phosphate, $\mathrm{Mg}_3\left(\mathrm{PO}_4\right)_2$ will contain 0.25 mole of oxygen atoms?

(A) 1.25×10^{-2}

(B) 2.5×10^{-2}

(C) 0.02

(D) 3.125×10^{-2}

Q.19

Haemoglobin contains 0.33% of iron by weight. The
molecular weight of haemoglobin is approximately 67200 . The number of iron atoms (at. wt. of $\mathrm{Fe} = 56$) present in one
molecule of haemoglobin is

(A) 6

(B) 1

(C) 2

(D) 4

Q.20

Identify the incorrect statement from the following

(A) Oxides of nitrogen in the atmosphere can cause the depletion of ozone layer

(B) Ozone absorbs the intense ultraviolet radiations of the sun

(C) Depletion of ozone layer is because of its chemical reactions with CFC

(D) Ozone absorbs infrared radiations

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Q.1 Answer:

1.568×10^4

Solution:

0.5% by weight means if Mol. wt. is 100 then mass of Se is 0.5. If at least one atom of Se is present in the molecule then

M. Wt $= \frac{100}{0.5} \times 78.4 = 1.568 \times 10^4$

Q.2 Answer:

6.023×10^{19}

Solution:

As we know that,

Density $= \frac{\text{mass}}{\text{volume}}$

$\Rightarrow$ **mass = density** $\times$ **volume**

Density of water $= 1\ \text{g/ml}$

Volume of water $= 0.0018\ \text{mL}$ **(Given)**

$\therefore$ **Mass of** $0.0018\ \text{mL}$ **water** $= 1 \times 0.0018 = 0.0018\text{gm}$

Molar mass of water $= 18\text{gm/mol}$

Number of moles in 0.0018gm **of water** $= \frac{\text{Given mass}}{\text{Molar mass}} = \frac{0.0018}{18} = 0.0001$ **mole**

As we know that,

Number of molecules of water in 1 mole $= 6.022 \times 10^{23}$ **molecules**

$\therefore$ **Number of molecules of water in** 0.0001 **mole** $= 6.022 \times 10^3 \times 0.0001 = 6.023 \times 10^{19}$

Hence, the number of water molecules present in a drop of water (volume $= 0.0018\text{ml}$**) at room temperature are** 6.023×10^{19}**.**

Q.3 Answer:

72

Solution:

Let mole fraction of O_2 **is x** $40 = 32 \times x + 80(1 - x)$

or $x = 5/6$

$a : b = x : (1 - x) = \frac{5}{6} : \frac{1}{6}$

When ratio is changed

$M_{\text{mixture}} = 32 \times \frac{1}{60} + 80 \times \frac{5}{6} = 72$

Q.4 Answer:

0.5×10^{-3}

Solution:

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Moles of H_2SO_4 in 98mg of H_2SO_4
$= \frac{1}{98} \times 0.098 = 0.001$
Moles of H_2SO_4 removes $= \frac{3.01 \times 10^{20}}{6.02 \times 10^{23}} = 0.5 \times 10^{-3} = 0.0005$
Moles of H_2SO_4 left $= 0.001 - 0.0005 = 0.5 \times 10^{-3}$

Q.5 Answer:

$II < I < III < IV$

Solution:

Mass of 6.023×10^{23} atoms of oxygen $= 16$ g Mass of one atom of oxygen
$= \frac{16}{6.023 \times 10^{23}} = 2.66 \times 10^{-23}$ g
Mass of 6.023×10^{23} atoms of nitrogen $= 14$ g
Mass of one atom of nitrogen
$= \frac{14}{6.023 \times 10^{23}} = 2.32 \times 10^{-23}$ g
Mass of 1 mole of oxygen $= 16$ g Mass of 1×10^{-10} mole of oxygen $= 16 \times 10^{-10}$ Mass of 1 mole of copper $= 63$ g Mass of 1×10^{-10} mole of copper $= 63 \times 1 \times 10^{-10} = 63 \times 10^{-10}$
So, the order of increasing mass is $II < I < III < IV$

Q.6 Answer:

43 g

Solution:

Moles of $C = n_{CO_2} = \frac{2.64}{44} = 0.06$
$\Rightarrow$ mass of $C = 0.72$
Mole of $H = 2 \times$ Moles of $H_2O =$
$2 \times \frac{1.26}{18} = 0.14 \Rightarrow$ mass of $H = 0.14$
Compound does not contains oxygen
So EF $\rightarrow C_{0.06}H_{0.14} \Rightarrow C_3H_7$
$\Rightarrow$ Lowest M.M. $= 43$

Q.7 Answer:

0.5

Solution:

Let weight of C be xg, then S will be $(14 - x)$g
$\frac{x/12}{(14-x)/32} = \frac{2}{1}$
$\therefore \quad x = 6$ g; Moles of $C = \frac{6}{12} = 0.5$

Q.8 Answer:

82.4

Solution:

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Let the alcohol be

ROH **and x its molecular weight**

$$\mathrm{ROH} + \mathrm{CH_3MgI} \rightarrow \mathrm{CH_4} + \mathrm{ROMgI}$$

$$\mathrm{xg} \qquad\qquad 16\mathrm{g}$$

$\frac{4.12}{1000}$ **g of alcohol will produce** $\frac{16}{x} \times \frac{4.12}{1000} g$ **of methane**

Methane actually obtained is $= \frac{16 \times 1.12}{22400}$ g

$\therefore \frac{16 \times 4.12}{x \times 1000} = \frac{16 \times 1.12}{22400} = x = 82.4$

Q.9 Answer:

0.060

Solution:

$$\underset{0.1}{4Ca_5(PO_4)_3F} + \overset{\text{L.R.}}{\underset{0.36}{18SiO_2}} + \underset{0.9}{30C} \longrightarrow 3P_4 + 2CaF_2 + 18CaSiO_3 + 30CO$$

18 moles of $\mathrm{SiO_2}$ **gives 3 moles of** $\mathrm{P_4}$

0.36 **moles of** $\mathrm{SiO_2}$ **will give**

$= \frac{3}{18} \times 0.36 = 0.06$ mole

Q.10 Answer:

84 g

Solution:

$\mathrm{CO} + \frac{1}{2}\mathrm{O_2} \rightarrow \mathrm{CO_2}$

$\mathrm{CO_2} + 2\mathrm{KOH} \rightarrow \mathrm{K_2CO_3} + \mathrm{H_2O}$

Moles of $\mathrm{KOH} = \frac{28}{56} = 0.50$

It corresponds to

0.25 mol **of** $\mathrm{CO_2}$

Hence mol of $\mathrm{CO} = 1 - 0.25 = 0.75 \equiv$ **mole of**

$\mathrm{CO_2}$ **formed Mol of KOH requred** $= 2 \times 0.75 = 1.5$

$= 1.5 \times 56 = 84$ g

Q.11 Answer:

0.6

Solution:

The required equation is $2\mathrm{KMnO_4} + 3\mathrm{H_2SO_4} \longrightarrow$

$\mathrm{K_2SO_4} + 2\mathrm{MnSO_4} + 3\mathrm{H_2O} + \quad 5[\mathrm{O}]$

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nascent oxygen

$2Fe\left(C_2O_4\right)+3H_2SO_4+3[O] \longrightarrow$
$Fe_2\left(SO_4\right)_3+2CO_2+3H_2O$

O required for 1 mol. of $Fe\left(C_2O_4\right)$ **is** $1.5, 5O$ **are obtained from 2 moles of** $KMnO_4$

$\therefore 1.5[O]$ **will be obtained from**

$=\frac{2}{5} \times 1.5=0.6$ **moles of** $KMnO_4$**.**

Q.12 Answer:

4

Solution:

Weight of Iron in 67200

$=\frac{0.33}{100} \times 67200=221.76$

Number of atoms of Iron

$=\frac{221.76}{56}=3.96 \equiv 4$

Q.13 Answer:

1.47×10^{22}

Solution:

Volume of gold foil $=25 \times 40 \times 0.25 \mathrm{~mm}^3$

$=250 \times 10^{-3} \mathrm{~cm}^3$

Mass of gold foil $=19.32 \times 250 \times 10^{-3} \mathrm{~g}$

$=4.83 \mathrm{~g}$

No. of gold atoms $=\frac{4.83}{197} \times N_A$

$=1.47 \times 10^{22}$

Q.14 Answer:

$0.1, 0.1$

Solution:

$I_2+2Cl_2 \longrightarrow ICl+ICl_3$

No. of moles	$\frac{25.4}{254}$	$\frac{14.2}{71}$	**0**	**0**
initially	**0.1**	**0.2**	**0**	**0**
No. of moles	**0**	**0**	**0.1**	**0.1**
after reaction				

Q.15 Answer:

$36 \mathrm{~g}$ of water

Solution:

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Number of molecules in 36 g of water
$= \frac{36}{18} \times 6.023 \times 10^{23} = 2 \times 6.023 \times 10^{23}$
Number of molecules in 28 g of CO
$= \frac{28}{28} \times 6.023 \times 10^{23} \quad = 1 \times 6.023 \times 10^{23}$
Number of molecules in 46 g of C_2H_5OH
$= \frac{46}{46} \times 6.023 \times 10^{23} = 1 \times 6.023 \times 10^{23}$
Number of molecules in 54 g of N_2O_5
$= \frac{54}{108} \times 6.023 \times 10^{23} = 0.5 \times 6.023 \times 10^{23}$
Hence, in these 36 g of water has largest number of molecules.

Q.16 Answer:

4 g of sulphur

Solution:

O atoms in 2g of oxygen
$= \frac{2 \times N_A}{16} = 0.125\ N_A$. 4 g $\left(= \frac{4}{32}\right.$ g **-atom** $\left.\right)$ **of sulphur also contains atom** $= 0.125\ N_A$

Q.17 Answer:

decrease twice

Solution:

Relative atomic mass =
Mass of one atom of the element
$1/12^{th}$ **part of the mass of one atom of Carbon** -12
or Mass of one atom of the element mass of one atom of the $C-12$ **Now if we use** $^1/_6$ **in place of** $1/_{12}$ **the formula becomes Relative atomic mass** $=$
$\frac{\text{Mass of one atom of element}}{\text{Mass of one atom of carbon}} \times 6$
$\therefore$ **Relative atomic mass decrease twice**

Q.18 Answer:

3.125×10^{-2}

Solution:

1 Mole of $Mg_3(PO_4)_2$ **contains 8 mole of oxygen atoms** $\therefore$ 8 **mole of oxygen atoms** $\equiv 1$ **mole of** $Mg_3(PO_4)_2$ **mole of** $Mg_3(PO_4)_2$
0.25 **mole of oxygen atom** $\equiv \frac{1}{8} \times 0.25$ **mole of** $Mg_3(PO_4)_2$
$= 3.125 \times 10^{-2}$ **mole of** $Mg_3(PO_4)_2$

Q.19 Answer:

4

Solution:

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Weight of Iron in $67200 = \frac{0.33}{100} \times 67200 = 221.76$
Number of atoms of Iron $= \frac{221.76}{56} = 3.96 \equiv 4$

Q.20 Answer:

Ozone absorbs infrared radiations

Solution:

Ozone doesn't absorb infrared radiation, it absorbs ultraviolet radiations.

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